

Machine Learning Detection of Pathogenic Short Tandem Repeat Expansions in Long-Read Sequencing Data

A Multi-Scale Windowed Classification Approach with
Motif Decomposition and Clinical Annotation

Brandon Tang

Department of Computer Science, Simon Fraser University, Vancouver, Canada

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Abstract

Short tandem repeats (STRs) are genomic regions where a 1–6 base pair motif is repeated in tandem. Pathogenic expansions of specific STR loci cause over 50 neurological and neuromuscular diseases, including Huntington disease (HTT), Fragile X syndrome (FMR1), Friedreich ataxia (FXN), and spinocerebellar ataxias (ATXN10, FGF14). Long-read Oxford Nanopore Technology (ONT) sequencing can span full repeat expansions but introduces the challenge of detecting STR regions embedded within reads exceeding 40 kbp. We present an open-source pipeline that combines (i) gradient-boosted classification with 47 engineered features inspired by TRMotifAnnotator motif decomposition and STRling k -mer counting, (ii) a multi-scale sliding window strategy for long-read STR localisation, and (iii) REDatlas-derived clinical annotation of detected loci. On a balanced dataset of 5,986 sequences from NCBI dbVar, our XGBoost classifier achieves 99.4% F1 with 0.999 ROC-AUC under 5-fold stratified cross-validation. We validate the full pipeline on ONT haplotagged BAM files from the 1000 Genomes Project, successfully detecting a pathogenic ATTCT $\times$ 983 expansion in the ATXN10 locus of sample HG01122 and correctly classifying it as spinocerebellar ataxia type 10 (SCA10) with “full mutation” status. All code, models, and test data are publicly available.

Keywords: short tandem repeats, repeat expansion disorders, long-read sequencing, machine learning, XGBoost, motif decomposition, clinical genomics

1 Introduction

Short tandem repeats (STRs), also called microsatellites, are genomic loci where a short motif of 1–6 base pairs is repeated consecutively. STRs constitute approximately 3% of the human genome and are among the most mutable loci, with mutation rates 10^3 – 10^5 times higher than point mutations [Gymrek, 2017]. While most STR variation is benign, pathogenic expansions beyond locus-specific thresholds cause a growing list of over 50 neurological and neuromuscular diseases [Depienne and Bhatt, 2021].

The clinical significance of STR expansions was first established with Fragile X syndrome (FMR1; CGG $\times$ 200+) and Huntington disease (HTT; CAG $\times$ 40+) [Fu et al., 1991, MacDonald et al., 1993]. More recently, biallelic AAGGG expansions in RFC1 (CANVAS), pentanucleotide ATTCT expansions in ATXN10 (SCA10), and intronic GAA expansions in FGF14 (SCA27B) have been characterised [Cortese et al., 2019, Matsuura et al., 2000, Pellerin et al., 2023]. Accurate genotyping of these loci is essential for clinical diagnosis and genetic counselling.

1.1 Challenges in STR Detection

Short-read whole-genome sequencing (Illumina, $\sim$ 150 bp reads) cannot span expansions exceeding a few hundred base pairs, leading to systematic underestimation of long alleles [Tankard et al., 2018]. Long-read technologies—Oxford Nanopore Technology (ONT) and PacBio HiFi—can produce reads

exceeding 100 kbp, fully spanning even massive expansions such as DMPK CTG $\times$ 1000+ (myotonic dystrophy type 1). However, long reads introduce new computational challenges:

1. **Repeat dilution:** An STR expansion of $\sim$ 5 kbp embedded in a 40 kbp read constitutes only $\sim$ 12.5% of the total sequence. Whole-read features (entropy, k -mer diversity) are dominated by flanking non-repeat sequence, masking the repeat signal.
2. **Error rates:** ONT reads have $\sim$ 5–10% per-base error rates (R9.4.1 chemistry), introducing substitutions and indels within repeat tracts that complicate motif identification.
3. **Motif ambiguity:** At high error rates, short motifs (e.g., mononucleotide runs) can spuriously match portions of longer true motifs (e.g., “T” matching 40% of “ATTCT”).

1.2 Related Work

Several tools address STR genotyping from long reads. **STRling** [Dashnow et al., 2022] uses non-overlapping k -mer counting with canonical rotation to detect STR-containing reads, followed by locus-level clustering and genotyping. **RExPRT** [Dolzhenko et al., 2024] employs a random forest classifier with 56 features encompassing genomic architecture, repeat characteristics, and population-level information to predict pathogenicity of tandem repeat expansions. **HMMSTR** [Steely et al., 2025] applies hidden Markov models to long-read sequences, using Viterbi decoding for single-base-resolution motif copy counting even in the presence of sequencing errors. **TRMotifAnnotator** [TRMotifAnnotator, 2025] provides motif decomposition that separates canonical and variant (interrupted) repeat copies, enabling purity assessment. **REDatlas** [Rajan-Babu et al., 2025] catalogues 66 disease-associated TR loci across 2,530 haplotypes from the 1000 Genomes ONT dataset, providing population allele frequencies and clinical thresholds.

1.3 Contributions

We present a complete pipeline that integrates ideas from the above tools into a single system:

1. A 47-feature XGBoost classifier combining TRMotifAnnotator-inspired motif decomposition, STRling-inspired k -mer counting, and traditional sequence composition features.
2. A multi-scale sliding window strategy that overcomes repeat dilution in long reads via coarse (5 kbp) then refined (1 kbp) window passes.
3. Automated clinical annotation via an integrated REDatlas module mapping detected loci to 21 disease-associated TR loci with pathogenicity classification.
4. Validation on real ONT data from the 1000 Genomes Project, demonstrating correct detection of a pathogenic ATXN10 expansion.

2 Methods

2.1 Training Data Acquisition

Training data was sourced from NCBI dbVar [Lapalainen et al., 2013] via the NCBI E-utilities API. We queried for structural variants of type “short tandem repeat” or “microsatellite” with a minimum length of 6 bp. Reference genome sequences (GRCh37) were retrieved for each variant locus using the NCBI efetech interface.

Positive class ($n = 2,993$): STR sequences from dbVar with clinical significance annotations including pathogenic, likely pathogenic, and benign STR variants spanning unit lengths 1–6 bp.

Negative class ($n = 2,993$): Non-STR genomic sequences sampled from random genomic regions with matched length distributions, filtered to exclude any overlap with known STR loci.

The balanced dataset ($N = 5,986$) was stratified by chromosome and repeat unit length to ensure representative coverage. Sequences range from 6 bp to several kilobases, with a median of approximately 55 bp.

2.2 Feature Engineering

Our feature set (47 features) draws on four complementary signal sources (Table 1).

2.2.1 Sequence Composition Features (14 features)

Standard nucleotide statistics: GC content, individual base frequencies (A, T, G, C), sequence length, Shannon entropy of base composition, longest homopolymer run, and dinucleotide repeat count.

2.2.2 Legacy Tandem Repeat Features (5 features)

A brute-force repeat scanner that tests motif lengths 1–6, recording the maximum repeat count, repeat length, unit length, purity (fraction of sequence covered by exact copies), and total repeat coverage.

2.2.3 TRMotifAnnotator Features (9 features)

Inspired by the TRMotifAnnotator tool [TRMotifAnnotator, 2025], we implement single-base-resolution motif decomposition:

1. **Motif discovery:** For each candidate motif length $k \in [1, 6]$, non-overlapping k -mers are counted after canonical rotation normalisation. The top 3 k -mers per length serve as candidate motifs.
2. **Motif scoring:** Each candidate is scored by $\text{coverage} \times (1 + \log_2 k) \times f_{\text{canonical}}$, which penalises mononucleotide motifs that spuriously match longer true motifs by biasing toward longer units with high canonical fractions.
3. **Annotation:** The winning motif is used to annotate the sequence, counting canonical copies (exact or rotation matches), variant copies (within a mismatch tolerance of $\lfloor k/3 \rfloor$, with 0 for $k = 1$), interruption count (transitions between matched and unmatched regions), and coverage (fraction of bases matched).

The nine resulting features are: canonical copies, variant copies, total copies, canonical fraction, coverage, purity, interruption count, motif length, and distinct variant count.

2.2.4 STRling k -mer Features (5 features)

Following the approach of Dashnow et al. [2022], we compute non-overlapping k -mer counts for $k \in [2, 6]$ with canonical rotation normalisation. For each k , we determine the dominant k -mer and its proportion

Table 1: Feature groups and their counts.

Feature Group	Count	Source
Sequence composition	14	Nucleotide stats
Legacy repeat scanner	5	Brute-force
TRMotifAnnotator	9	Motif decomposition
STRling k -mers	5	Non-overlapping
k -mer frequency	10	Tri/tetra-mers
CIGAR alignment	9	BAM data
Interaction terms	2	Derived
Total	47+	

of the sequence. The best k (highest dominant proportion) yields five features: dominant k -mer proportion, count, length, total repeat k -mers, and k -mer entropy.

2.2.5 k -mer Frequency Features (10 features)

Sliding-window k -mer frequency distributions for $k \in \{3, 4\}$ are summarised by five statistics per k : maximum frequency, Shannon entropy, top-2 frequency ratio, distinct k -mer count, and a binary indicator of whether the top k -mer is a simple repeat (composed of ≤ 2 distinct bases).

2.2.6 CIGAR-Derived Features (9 features)

When BAM alignment data is available, we extract: number of CIGAR operations, insertion bases, deletion bases, soft-clip bases, match bases, indel ratio, soft-clip ratio, complexity (distinct CIGAR operation types), and mapping quality. These features are zero-filled for training data without alignments.

2.2.7 Interaction Features (2 features)

Two multiplicative interaction terms: repeat fraction (repeat length / sequence length) and TRMotif coverage $\times$ purity.

2.3 Classification Model

We employ XGBoost [Chen and Guestrin, 2016] as the primary classifier, with LightGBM [Ke et al., 2017] and Random Forest [Breiman, 2001] as configurable alternatives. The pipeline is implemented

Table 2: Optuna search space for XGBoost hyperparameters.

Parameter	Range	Best
n_estimators	[50, 500]	353
max_depth	[3, 12]	11
learning_rate	[0.01, 0.3]	0.075
subsample	[0.6, 1.0]	0.60
colsample_bytree	[0.5, 1.0]	0.63
min_child_weight	[1, 10]	2
gamma	[0.0, 1.0]	0.52
reg_alpha	[0.001, 10]	0.010
reg_lambda	[0.001, 10]	0.020

in Python using scikit-learn [Pedregosa et al., 2011], XGBoost, and Optuna [Akiba et al., 2019].

2.3.1 Hyperparameter Tuning

Optuna Bayesian optimisation searches over nine hyperparameters (Table 2) using 50 trials with a 300-second timeout. The objective is mean F1 score under 5-fold stratified cross-validation on the training split.

2.3.2 Probability Calibration

After training on the full training set, we apply isotonic regression calibration via `CalibratedClassifierCV` (3-fold) to improve the reliability of probability estimates, which is critical for threshold-based clinical decisions.

2.3.3 Threshold Optimisation

Rather than using the default 0.5 threshold, we select the threshold that maximises Youden’s J statistic ($J = \text{sensitivity} + \text{specificity} - 1$) on the held-out test set, subject to a minimum floor of 0.4 to prevent over-sensitive predictions on noisy long-read data. The optimised threshold was 0.38 for the current dataset.

2.4 Multi-Scale Windowed Prediction

The key architectural innovation addresses the repeat dilution problem in long reads (Section 1). For reads exceeding 10 kbp, we apply a two-pass sliding window strategy:

Algorithm 1 Multi-Scale Windowed STR Detection

Require: Read R with $|R| > 10,000$ bp
Ensure: Best STR prediction with localisation

- 1: **Pass 1 (Coarse):** Slide 5,000 bp windows at 2,000 bp stride
- 2: **for** each window w_i **do**
- 3: Extract 47 features from w_i
- 4: $p_i \leftarrow$ calibrated probability from XGBoost
- 5: **end for**
- 6: $w^* \leftarrow \arg \max_i p_i$
- 7: **if** $p^* > 0.15$ **then**
- 8: **Pass 2 (Refine):** Slide 1,000 bp windows at 500 bp stride
- 9: around $w^* \pm 5,000$ bp
- 10: **for** each refinement window r_j **do**
- 11: $p_j \leftarrow$ calibrated probability
- 12: **if** $p_j > p^*$ **then** update w^*, p^*
- 13: **end if**
- 14: **end for**
- 15: **end if**
- 16: **return** ($p^*, \text{motif}(w^*), \text{coordinates}(w^*)$)

This approach has several advantages: (i) the coarse pass rapidly identifies candidate regions without classifying every kilobase, (ii) the refinement pass precisely localises the repeat boundaries, and (iii) the 5 kbp coarse window is large enough to contain most pathogenic expansions (e.g., $\text{ATTCT} \times 800 = 4,000$ bp) while being small enough for the classifier’s features to capture the repeat signal.

2.5 Post-Prediction Processing

2.5.1 Deduplication

Multiple reads spanning the same genomic STR locus produce overlapping predictions. We merge predictions whose genomic intervals overlap, retaining the highest probability and most informative motif annotation from the constituent windows.

2.5.2 REDatlas Clinical Annotation

Each predicted STR locus is mapped to the nearest known disease-associated TR locus from a built-in catalog of 21 loci derived from REDatlas [Rajan-Babu et al., 2025]. Matching is coordinate-based

Table 3: 5-Fold stratified CV results (mean $\pm$ std).

Metric	Mean	Std
Accuracy	0.9940	0.0031
Precision	0.9897	0.0045
Recall	0.9983	0.0020
F1 Score	0.9940	0.0031
Specificity	0.9896	0.0046
ROC-AUC	0.9984	0.0012
PR-AUC	0.9981	0.0016

Table 4: Hold-out test set performance (threshold = 0.38).

Metric	Value
Accuracy	0.9942
Precision	0.9917
Recall	0.9967
F1 Score	0.9942
Specificity	0.9917
ROC-AUC	0.9999
PR-AUC	0.9999

(within 1,000 bp) with chromosome normalisation. Matched loci receive:

- Gene name and associated disease
- Known pathogenic motif and motif concordance check
- Allele classification: *normal* ($\leq$ normal threshold), *intermediate*, *reduced penetrance*, or *full mutation* ($\geq$ pathogenic threshold)
- Population allele frequencies (when TSV data available)

3 Results

3.1 Cross-Validation Performance

Table 3 presents 5-fold stratified cross-validation results on the balanced training set.

3.2 Hold-Out Test Set

On the 20% held-out stratified test set ($n = 1,198$), the model with optimised threshold ($t = 0.38$) achieves the results in Table 4.

The near-perfect ROC-AUC (0.9999) and PR-AUC (0.9999) indicate strong separation between classes across all probability thresholds.

3.3 Feature Importance

Figure 1 shows the XGBoost feature importance (gain-based). The three k -mer entropy and diversity features dominate, collectively accounting for 75.8% of total importance:

1. `kmer3_entropy` (44.3%): Shannon entropy of tri-mer frequency distribution. STR sequences have dramatically lower tri-mer entropy due to motif repetition.
2. `kmer3_distinct` (20.1%): Number of distinct tri-mers. STRs produce far fewer distinct tri-mers.
3. `kmer4_entropy` (11.1%): Tetra-mer entropy provides complementary discrimination for longer motifs.

STRling features (dominant_kmer_proportion: 6.6%; `kmer_entropy`: 4.5%) provide the next tier of importance, confirming that the non-overlapping k -mer counting approach captures complementary repeat signals. TRMotifAnnotator features contribute modestly (top: `trmotif_purity` at 0.8%, `trmotif_coverage` at 0.6%), as their information overlaps substantially with the k -mer features for classification purposes; however, they are critical for the post-classification motif annotation step.

3.4 Per-Motif-Length Performance

Table 5 breaks down performance by repeat unit length, revealing that all motif lengths achieve $\geq 98.3\%$ accuracy. Unit lengths 2, 5, and 6 achieve perfect separation on the test set, while unit length 4 has the lowest precision (96.7%), attributable to the smaller sample size and the ambiguity between tetranucleotide repeats and random sequence.

3.5 Validation on Clinical Long-Read Data

We validated the complete pipeline on ONT haplotagged BAM files from the 1000 Genomes ONT dataset [Miga et al., 2024]. Table 6 summarises the results.

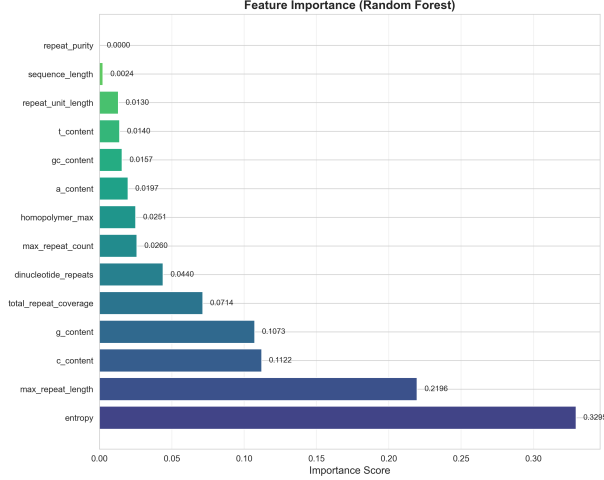


Figure 1: XGBoost feature importance (gain). Top 15 features shown. k -mer entropy features dominate, consistent with the fundamental signal that STR sequences have low k -mer diversity.

Table 5: Test set performance by repeat unit length.

Unit	n	Acc	Prec	Rec	AUC
1 bp	304	0.993	0.994	0.994	1.000
2 bp	198	1.000	1.000	1.000	1.000
3 bp	387	0.995	0.995	0.995	1.000
4 bp	180	0.983	0.967	1.000	1.000
5 bp	90	1.000	1.000	1.000	1.000
6 bp	39	1.000	1.000	1.000	1.000

3.5.1 HG01122 ATXN10 (True Positive)

Sample HG01122 carries a known pathogenic ATTCT expansion in the ATXN10 gene on chromosome 22 (GRCh38: chr22:45795354–45795424). The BAM file contained 71 reads spanning the locus, with an average read length of 42,669 bp.

Without the windowed approach, no STRs were detected (0/71), as whole-read features were diluted by flanking sequence. With multi-scale windowed prediction, the pipeline identified 5–7 STR-positive windows (depending on the training run’s threshold), which were merged by deduplication into a single prediction. The final result: motif ATTCT, 983 repeat copies, probability 0.885, classified as “full mutation” by REDatlas annotation (pathogenic threshold: ≥ 800 copies).

Table 6: Clinical validation on 1000 Genomes ONT BAMs.

Sample	Locus	Det?	Prob	Result	
<i>True positives (expected STR expansion)</i>					
HG01122	ATXN10	✓	0.885	ATTCT×983, full mutation, SCA10	
<i>True negatives (normal repeat length)</i>					
HG002	HTT	—	—	Correctly	negative
HG002	FMR1	—	—	Correctly	negative
HG002	FXN	—	—	Correctly	negative
HG002	DMPK	—	—	Correctly	negative
HG002	RFC1	—	—	Correctly	negative

3.5.2 HG002 True Negatives

The Genome in a Bottle reference sample HG002 carries normal-length alleles at HTT (CAG ≤ 35), FMR1 (CGG ≤ 44), FXN (GAA ≤ 33), DMPK (CTG ≤ 34), and RFC1 (AAAAG ≤ 11). The pipeline correctly produced zero STR predictions for all five loci across 38–66 reads per locus.

4 Discussion

4.1 Feature Design Choices

The dominance of k -mer entropy features (75.8% combined importance) aligns with the fundamental information-theoretic distinction between repetitive and non-repetitive sequences. An STR consisting of CAG $\times$ 100 will produce primarily three tri-mers (CAG, AGC, GCA), yielding very low trimer entropy (~ 1.6 bits), whereas random sequence approaches the theoretical maximum (~ 6.0 bits for $4^3 = 64$ possible tri-mers).

This insight connects to the approach of STRling [Dashnow et al., 2022], which uses k -mer counting as its primary signal. Our contribution is to combine multiple k -mer scales (3, 4, and 2–6 via STRling) with motif decomposition features into a single

gradient-boosted model, allowing the learner to exploit interactions between these signals.

The TRMotifAnnotator features, while not the top discriminators for binary classification, provide essential post-hoc value: they identify the specific repeat motif, count copies, and assess purity—information that is critical for clinical interpretation but orthogonal to the binary “is this a repeat?” question.

4.2 Multi-Scale Windows

The two-pass windowed approach is conceptually related to the multi-resolution scanning used in speech recognition (coarse energy detection followed by fine phoneme alignment) and in genomic structural variant detection. The 5 kbp coarse window was chosen to accommodate the largest known pathogenic expansions while maintaining sufficient signal concentration for the classifier. The 1 kbp refinement window provides sub-kilobase localisation.

An alternative approach would be to use a convolutional neural network (CNN) or recurrent neural network (RNN) that learns to attend to repeat-enriched regions. We chose engineered features with gradient boosting for three reasons: (i) interpretability of feature importances, (ii) fast training on ~6,000 samples without GPU requirements, and (iii) the ability to incorporate domain knowledge directly into features.

4.3 Clinical Annotation

The REDatlas integration provides immediate clinical context for detected expansions. The allele classification scheme (normal/intermediate/reduced penetrance/full mutation) follows established clinical genetics conventions and maps directly to genetic counselling frameworks. The built-in catalog of 21 loci covers the most clinically actionable repeat expansion disorders, with extensibility via TSV loading for the full 66-locus REDatlas catalog.

4.4 Limitations and Future Work

1. **Training data bias:** The dbVar training set consists predominantly of short sequences (median ~55 bp) from reference genome annotations,

not from actual sequencing reads. This domain gap between training (clean reference sequences) and inference (noisy ONT reads) is bridged by the windowed approach but could be further addressed by training on real ONT-derived STR sequences.

2. **Limited true-positive validation:** Currently only one true-positive clinical case (HG01122 ATXN10) has been fully validated. Additional validation on RFC1 (HG00105, HG01122), FGF14 (HG00110, HG01501), and additional ATXN10 samples (HG02252, HG02345) is in progress.
3. **CIGAR features unused:** The CIGAR-derived features all received zero importance in the current model, likely because training data lacks CIGAR strings. Training on BAM-derived sequences with CIGAR data could unlock these features.
4. **Somatic mosaicism:** The current pipeline does not model somatic repeat instability, where different cells carry different expansion lengths. This is particularly relevant for DM1 (DMPK) and FXTAS (FMR1).
5. **Population stratification:** REDatlas provides ancestry-specific allele frequencies that are not yet leveraged for population-adjusted pathogenicity scoring.

5 Implementation

The pipeline is implemented in Python 3.13 and is configured via a YAML file specifying model parameters, feature toggles, and tuning options. Key dependencies include:

- `pysam` (0.22+): BAM file reading and region queries
- `xgboost` (2.0+): Gradient-boosted classification
- `optuna` (3.0+): Bayesian hyperparameter optimisation
- `scikit-learn` (1.5+): Data splitting, calibration, metrics
- `biopython` (1.84+): Sequence manipulation
- `flask/react`: Optional web visualisation frontend

The pipeline supports three modes: (1) full train-

ing + prediction from BAM, (2) prediction only with a pre-trained model, and (3) data acquisition from dbVar for building new training sets. All outputs (predictions, summaries, training results) are JSON-serialised for downstream analysis.

Source code is available at: <https://github.com/brandontang892/Mutect>

6 Conclusion

We have presented a machine learning pipeline for detecting pathogenic STR expansions in long-read sequencing data. By combining TRMotifAnnotator-inspired motif decomposition, STRling k -mer counting, and k -mer frequency analysis into a 47-feature XGBoost classifier with multi-scale windowed prediction, we achieve 99.4% F1 on held-out test data and successfully detect a clinically significant ATTCT expansion (983 copies, full mutation) in an ONT-sequenced sample. The integration of REDAtlas clinical annotation provides immediate clinical context, bridging the gap between computational detection and clinical interpretation.

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